EE1003
Introduction to Computer I

PA1: Dinosaur Game

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Slides Credit: TA Ting-Yun Liu

Problem

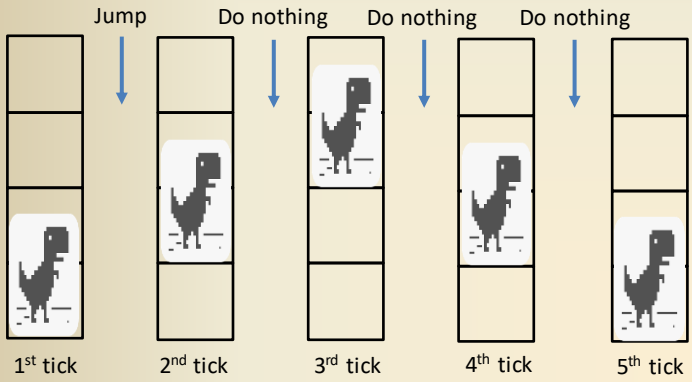
- This assignment requires you to complete a Dinosaur Game usually seen on Chrome.
  - Initial input: A 8-digit number represents the obstacle distribution. No fool-proofing.
  - Every tick input: 1-digit number represents obstacle next tick, no fool-proofing. 1-digit number represents the dinosaur movement, NEED fool-proofing.
  - Output: Map of Dinosaur Game.

Movement Command

- The scene goes left, the dinosaur stays.
- 3 types of dinosaur movements, calls *command*. Need fool-proofing.
  - Jump: Position at y=1,2,1,0 for next four ticks.
  - Crouch: Height=1, force to y=0.
  - Do nothing: Straighten to weight=2 if it was crouching; Keep on jumping procedure if it was jumping.

Movement Command

- Jump

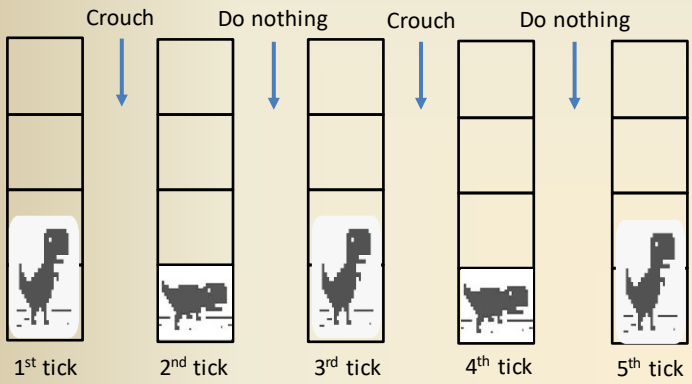


Movement Command

- The scene goes left, the dinosaur stays.
- 3 types of dinosaur movements, calls *command*. Need fool-proofing.
  - Jump: Position at y=1,2,1,0 for next four ticks.
  - Crouch: Height=1, force to y=0.
  - Do nothing: Straighten to weight=2 if it was crouching; Keep on jumping procedure if it was jumping.

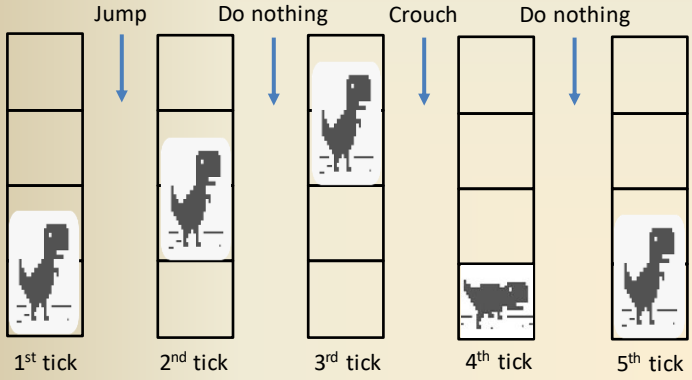
Movement Command

- Crouch



Movement Command

◆Jump and crouch



Obstacle Code

◆Four types of obstacles need to avoid.

(a)Cactus, height=1, grow from y=0.

(b)Cactus, height=2, grow from y=0.

(c)Pterosaur, flying at y=1.

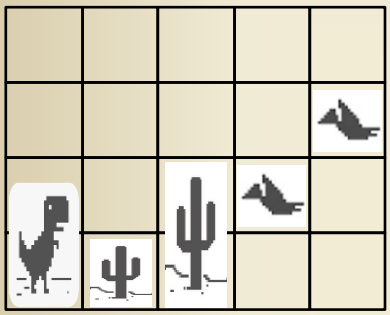
(d)Pterosaur, flying at y=2.

◆Obstacles will be given in coded format.

Obstacle	(a)	(b)	(c)	(d)
Obstacle code	1	2	3	4

Obstacle Code

- Picture of obstacle types



(a)Cactus, height=1, grow from y=0.
(b)Cactus, height=2, grow from y=0.
(c)Pterosaur, flying at y=1.
(d)Pterosaur, flying at y=2.

Map Representation

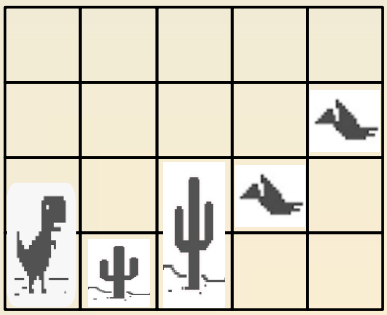
- Map shows the scene of Dinosaur Game.
- Characters to represent the scene, calls *symbol*.
  - A space before symbols is needed.

Object	Dinosaur	Height=1 Cactus	Height=2 Cactus	Y=1 Pterosaur	Y=2 Pterosaur
Char	' D'	' C'	' C'	' P'	' P'

Output Screen

◆Example of coded output screen

```
|   |   |   |   |   |
|   |   |   |   | P |
| D |   | C | P |   |
| D | C | C |   |   |
```



Output Screen

```
=====Welcome to the Dinosaur Game=====
Please enter the initial obstacle setup of the Dinosaur Game.
Enter -1 to quit: 01003000
|   |   |   |   |   |   |   |   |   |
|   |   |   |   |   |   |   |   |   |
| D |   |   |   | P |   |   |   |   |
| D |   | C |   |   |   |   |   |   |
Please enter a command to control the dinosaur, 0 for no
action, 1=jump, 2=crouch: 0
Please enter the next round's obstacle setup of the Dinosaur
Game. Enter -1 to quit: 1
|   |   |   |   |   |   |   |   |   |
|   |   |   |   |   |   |   |   |   |
| D |   |   |   | P |   |   |   |   |
| D | C |   |   |   |   |   |   | C |
```

Output Screen

```
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's obstacle setup of the Dinosaur Game. Enter
-1 to quit: 2
|  |  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  |  |  |
| D|  |  | P|  |  |  |  | C|
| C|  |  |  |  |  |  | C| C|
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
The dinosaur cannot jump again, please enter another command: 3
The dinosaur doesn't understand this command, try another one: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter
-1 to quit: 3
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  | P|  |  |  |  | C| P|
| D|  |  |  |  |  | C| C|  |
```


Output Screen

```
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter
-1 to quit: 4
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  | P|
|  | P|  |  |  |  | C| P|  |
| D|  |  |  |  | C| C|  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter
-1 to quit: 0
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  | P|  |
| P|  |  |  |  | C| P|  |  |
| D|  |  |  | C| C|  |  |  |
```

Output Screen

```
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's obstacle setup of the Dinosaur Game. Enter
-1 to quit: 0
|  |  |  |  |  |  |  |  |  |
| D|  |  |  |  |  | P|  |  |
| D|  |  |  | C| P|  |  |  |
|  |  |  | C| C|  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 0
Please enter the next round's obstacle setup of the Dinosaur Game. Enter
-1 to quit: 0
| D|  |  |  |  |  |  |  |  |
| D|  |  |  |  | P|  |  |  |
|  |  |  | C| P|  |  |  |  |
|  |  | C| C|  |  |  |  |  |
```


Output Screen

```
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter
-1 to quit: 0
|  |  |  |  |  |  |  |  |  |
|  |  |  |  | P|  |  |  |  |
|  |  | C| P|  |  |  |  |  |
| D| C| C|  |  |  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump,
2=crouch: 1
Please enter the next round's obstacle setup of the Dinosaur Game. Enter
-1 to quit: 0
|  |  |  |  |  |  |  |  |  |
| D|  |  | P|  |  |  |  |  |
| D| C| P|  |  |  |  |  |  |
| C| C|  |  |  |  |  |  |  |
```

Output Screen

```
Please enter a command to control the dinosaur, 0 for no action, 1=jump, 2=crouch: 0
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1 to quit:
0
| D|  |  |  |  |  |  |  |
| D|  | P|  |  |  |  |  |
| C| P|  |  |  |  |  |  |
| C|  |  |  |  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump, 2=crouch: 2
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1 to quit:
0
|  |  |  |  |  |  |  |  |
|  | P|  |  |  |  |  |  |
| P|  |  |  |  |  |  |  |
| D|  |  |  |  |  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump, 2=crouch: 1
Please enter the next round's obstacle setup of the Dinosaur Game. Enter -1 to quit:
0
What a pity, it was close :(
=====Thanks for playing Dinosaur Game, see you next time=====
```


Bonus Part 1

- Double Layers Dinosaur Game:
  - 2 layers of original Dinosaur Game
  - 1 ceiling that allows the dinosaur to stand on, to cross through, or blocks the dinosaur from jumping.
  - Double jump is allowed under certain circumstance.

Bonus Part 1

- Movement command remain unchanged.
- Input obstacle code has two parts.
- 1 more ceiling distributing information is given.

Bonus Part 1 Output Screen

```
======Welcome to the Dinosaur Game=====
Please enter the first layer initial obstacle setup of the Dinosaur
Game. Enter -1 to quit: 01234023
Please enter the second layer initial obstacle setup of the Dinosaur
Game: 00000000
Please enter the initial distribution of gap in the ceiling, 0 for
gap, 1 for complete ceiling: 000000000
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  =  =  =  =
|  |  |  |  |  |  |  |  |  |
|  |  |  |  |  | P|  |  |  |
| D|  |  | C| P|  |  | C| P|
| D|  | C| C|  |  |  | C|  |
```

Bonus Part 1 Output Screen

```
Please enter a command to control the dinosaur, 0 for no action,
1=jump, 2=crouch: 0
Please enter the next round's first layer obstacle setup of the
Dinosaur Game. Enter -1 to quit: 1
Please enter the next round's second layer obstacle setup of the
Dinosaur Game: 0
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  =  =  ====
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  P|  |  |  |  |  |
| D|  | C| P|  |  | C| P|  |  |
| D| C| C|  |  |  | C|  | C|  |
```


Bonus Part 1 Output Screen

```
Please enter a command to control the dinosaur, 0 for no action,
1=jump, 2=crouch: 1
Please enter the next round's first layer obstacle setup of the
Dinosaur Game. Enter -1 to quit: 2
Please enter the next round's second layer obstacle setup of the
Dinosaur Game: 4
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  P|  |
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  =  =  ======
|  |  |  |  |  |  |  |  |  |  |
| D|  |  | P|  |  |  |  |  |  |
| D| C| P|  |  | C| P|  | C|  |
| C| C|  |  |  | C|  | C| C|  |
```

Bonus Part 1 Output Screen

```
Please enter a command to control the dinosaur, 0 for no action,
1=jump, 2=crouch: 1
The dinosaur cannot jump again, please enter another command: 0
Please enter the next round's first layer obstacle setup of the
Dinosaur Game. Enter -1 to quit: 4
Please enter the next round's second layer obstacle setup of the
Dinosaur Game: 4
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  | P| P|
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |  |  |
=  =  =  =  =  =  ========
| D|  |  |  |  |  |  |  |  |  |
| D|  | P|  |  |  |  |  | P|  |
| C| P|  |  | C| P|  | C|  |  |
| C|  |  |  | C|  | C| C|  |  |
```


Bonus Part 1 Output Screen

```
Please enter a command to control the dinosaur, 0 for no action,
1=jump, 2=crouch: 1
Please enter the next round's first layer obstacle setup of the
Dinosaur Game. Enter -1 to quit: 1
Please enter the next round's second layer obstacle setup of the
Dinosaur Game: 1
Please enter the next round's ceiling or gap status: 0
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  | P| P|  |
|  |  |  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  |  | C|
=  =  =  =  =  ========  =
| D|  |  |  |  |  |  |  |  |
|  | P|  |  |  |  |  | P|  |
| P|  |  | C| P|  | C|  |  |
|  |  |  | C|  | C| C|  | C|
```

Bonus Part 1 Output Screen

```
Please enter a command to control the dinosaur, 0 for no action,
1=jump, 2=crouch: 1
The dinosaur cannot jump again, please enter another command: 0
Please enter the next round's first layer obstacle setup of the
Dinosaur Game. Enter -1 to quit: 0
Please enter the next round's second layer obstacle setup of the
Dinosaur Game: 0
Please enter the next round's ceiling or gap status: 1
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  | P| P|  |  |
| D|  |  |  |  |  |  |  |  |  |
| D|  |  |  |  |  |  | C|  |  |
=  =  =  =  =======  ====
|  |  |  |  |  |  |  |  |  |  |
| P|  |  |  |  |  |  | P|  |  |
|  |  |  | C| P|  | C|  |  |  |
|  |  |  | C|  | C| C|  | C|  |
```


Bonus Part 1 Output Screen

```
Please enter a command to control the dinosaur, 0 for no action, 1=jump, 2=crouch: 2
Please enter the next round's first layer obstacle setup of the Dinosaur Game. Enter
-1 to quit: 0
Please enter the next round's second layer obstacle setup of the Dinosaur Game: 0
Please enter the next round's ceiling or gap status: 0
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  | P| P|  |  |  |
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  | C|  |  |  |
=  =  =  =======  ===  =
|  |  |  |  |  |  |  |  |  |  |
|  |  |  |  |  |  | P|  |  |  |
|  |  | C| P|  | C|  |  |  |  |
| D| C|  | C| C|  | C|  |  |  |
Please enter a command to control the dinosaur, 0 for no action, 1=jump, 2=crouch: 2
Please enter the next round's first layer obstacle setup of the Dinosaur Game. Enter
-1 to quit: -1
Congratulations, you WIN the game!!!
=====Thanks for playing Dinosaur Game, see you next time=====
```

Bonus Part 2

- Automatically pass the game.
- Solution is not unique and all accept if legal.
- No fool-proofing is needed and is not allowed to enter illegal command.
- No sample will show since the answer is not unique.



Report

- How to compile and execute your program; (You can use a screenshot to explain. In this PA, it can be done in one click)
- The degree of completion of the assignment; (If you complete all requirements, just specify all)
- The hardness of this assignment and how you overcome it;
- Bonus function(s) you implement if any;
- Any suggestions about this programming assignment?
- You can also put anything related to the PA in your report, such as pseudocode, control flow diagram, programming development thoughts, etc.



Correctness

- Your program for basic and bonus part 1 will be judged by our On-line Judge (OJ) system and the bonus part 2 will be judged by checker. The machine information is as follows:
  - OS: Ubuntu 20.04 LTS
  - OJ Version：Hydro OJ v2.33.13
  - Compiler：GNU GCC 9.3.0
  - Supporting C++ version：C++ 11





Q&A

- If you have any other problems, you can also send an email to me.
- TA email is tim0974040@gmail.com
- Please cc andyygchen@ee.ncu.edu.tw





EE1003
Introduction to Computer I

Thank you!



